

This project aims to investigate **architecture-level processor power modeling** using deep neural networks, with the goal of learning a fast and accurate surrogate model that predicts CPU power directly from microarchitectural configuration parameters and runtime event statistics. This problem is of substantial importance because accurate power estimation typically requires RTL implementation, logic synthesis, and signoff-grade power analysis, which are too costly for early-stage design-space exploration, while classical analytical tools such as McPAT often exhibit large modeling error on modern processors [1, 2].

To provide context and technical grounding, we will study the ArchPower paper as the primary reference and examine representative prior work on analytical, calibrated, and hybrid machine-learning-based power modeling, including **McPAT**, **McPAT-Calib**, **PANDA**, and recent neural-network-based approaches for microarchitectural power estimation [1–5].

Our experiments will use the open-source **ArchPower dataset**, which contains 200 samples collected from 25 CPU configurations and 8 workloads, with 101 input features per sample and fine-grained labels for total power, 11 component-level powers, and four power groups [1]. No new data collection is required; instead, we will use the official dataset and its predefined Balance, Small, and Large train/test scenarios to evaluate both in-distribution accuracy and cross-scale generalization [1].

Methodologically, we propose a **multi-task DNN for tabular regression**, implemented as a shared feature encoder with separate prediction heads for total power and component-level power, so that the model can exploit the hierarchical supervision provided by ArchPower. We will reproduce strong baselines such as McPAT-Calib and PANDA from the available implementations [1, 3, 4], and then investigate improvements through normalization, dropout, residual MLP blocks, loss reweighting, and feature grouping by CPU component.

Evaluation will follow prior work using **MAPE** and **correlation coefficient R** [1], and will be supplemented by MAE/RMSE, mean and standard deviation over multiple random seeds, predicted-versus-ground-truth scatter plots, and component-wise error visualizations; we expect the proposed DNN to outperform single-output baselines in the balanced setting while also clarifying the remaining challenge of out-of-distribution generalization [1].

References

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